

Quantum

Lecture 12

Development: Light as a wave or Particle

- Young's Experiment: The light behaves as a wave.
- Photoelectric effect: The light behaves as a particle. These particles are called photons and they are associated with momentum.
- Oops.....Isn't mass of light is zero? That will imply a zero momentum.
- Einstein: This is not true when you travel with the velocity of light.

Momentum of a photon

$$E=mc^2 = \sqrt{p^2c^2 + m_o^2c^4}$$

$$E=h\nu = \frac{hc}{\lambda}$$

Rest mass=0

$$\frac{h}{\lambda} = \frac{E}{c}$$

$$p = \frac{E}{c}$$

$$\lambda = \frac{h}{p}$$

Light can behave as wave or particle depending on the experiment, and photons can have momentum.

Difference between a baseball and an electron

- Classical mechanics describes the motion of a baseball, the spinning of a top, and the flight of an airplane.
- Quantum mechanics describes the motion of electrons and the shapes of molecules such as trans fats, as well as electrical conductivity and superconductivity.

Schrödinger Equation



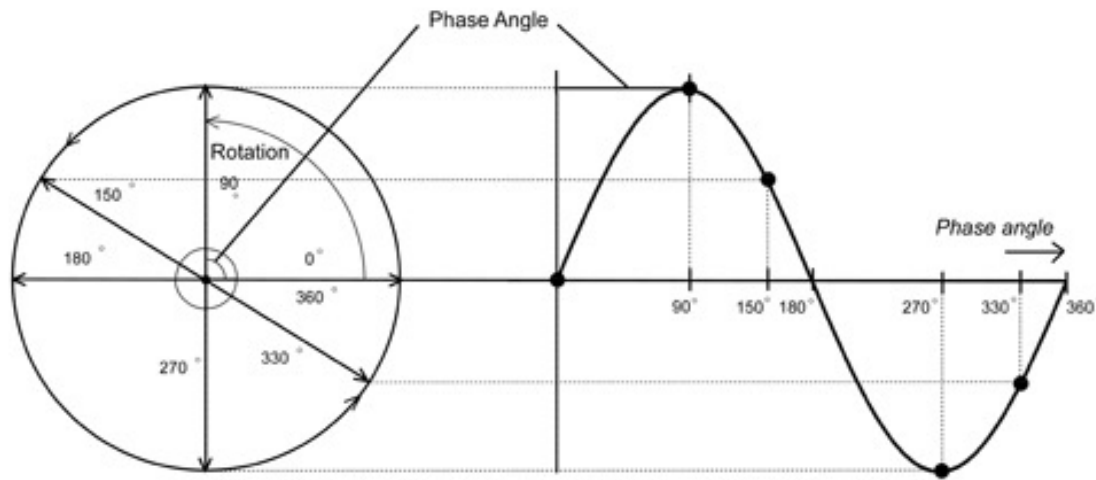
Basis of Schrödinger Equation

- Electrons behaves like wave.
- Momentum of electron is related to the wavelength by equation

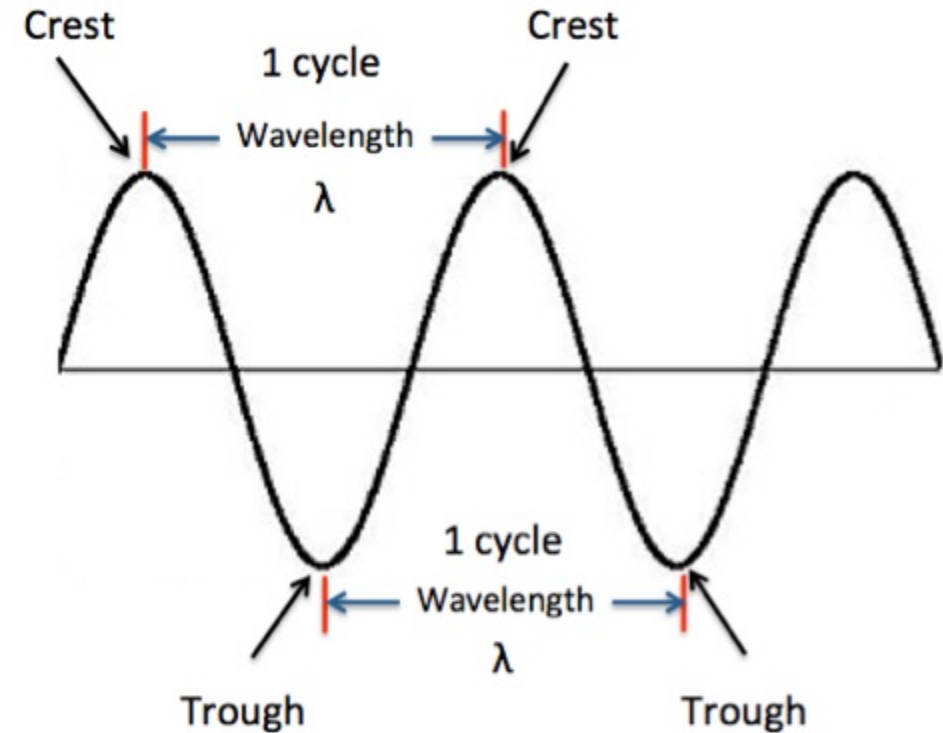
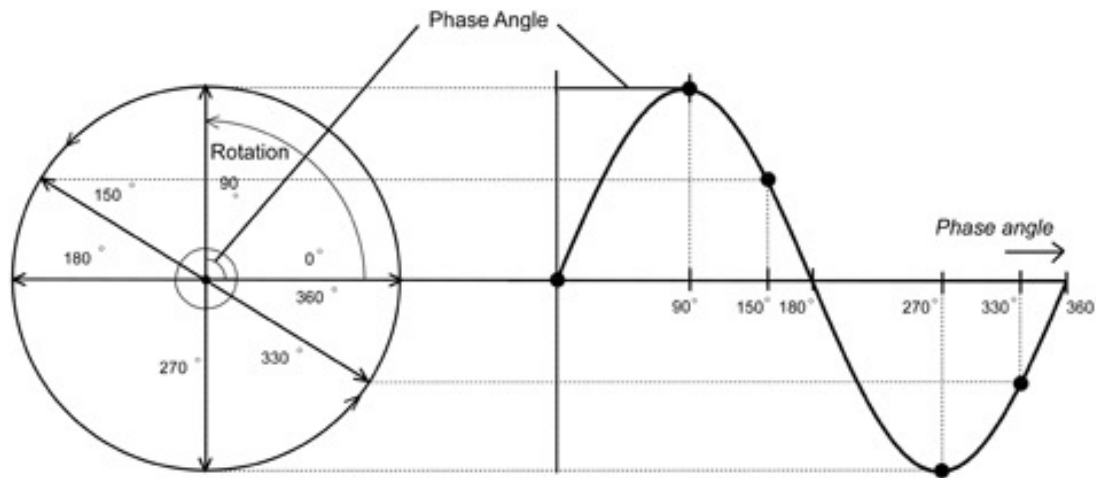
$$\lambda = \frac{h}{p}$$

Equation of a standing wave

$$y(\theta) = A \sin \theta$$



One cycle corresponds to 2π rotation and 1λ translation



Crest, Trough and Wavelength

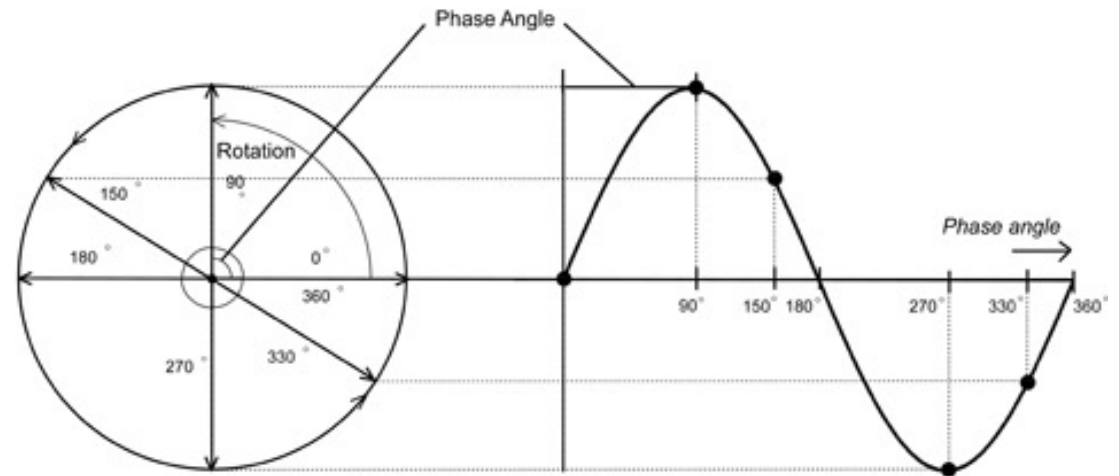
Equation of a standing wave

$$y(\theta) = A \sin \theta$$

$$\frac{\theta}{x} = \frac{2\pi}{\lambda}$$

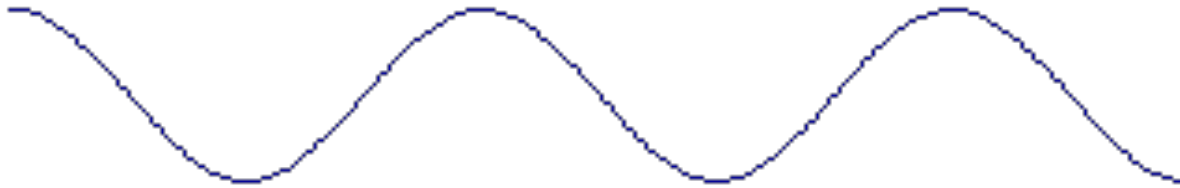
$$\theta = \frac{2\pi}{\lambda} x$$

$$y(x) = A \sin 2\pi \frac{x}{\lambda}$$



Equation of a wave

- If wave travels in positive x-direction with velocity v



Distance travelled by the wave in time $t = v \cdot t$

$$y(x, t) = A \sin 2\pi \left(\frac{x - vt}{\lambda} \right)$$

How amplitude changes with
 x

Time-independent Differential Equation of a wave

$$y(x, t) = A \sin 2\pi \left(\frac{x}{\lambda} - \frac{vt}{\lambda} \right)$$

$$\frac{dy}{dx} = -A \left(\frac{2\pi}{\lambda} \right) \cos 2\pi \left(\frac{x}{\lambda} - \frac{vt}{\lambda} \right)$$

$$\frac{d^2 y}{dx^2} = -A \left(\frac{2\pi}{\lambda} \right)^2 \sin 2\pi \left(\frac{x}{\lambda} - \frac{vt}{\lambda} \right)$$

$$\frac{d^2 y(x)}{dx^2} = - \left(\frac{2\pi}{\lambda} \right)^2 y(x)$$

Differential Equation of a wave associated with electron

- The de Broglie wave for a particle is made up of a superposition of an infinitely large number of waves of the form

$$\psi(x, t) = A \sin 2\pi \left(\frac{x}{\lambda} - \frac{vt}{\lambda} \right)$$

$$\frac{d^2\psi(x)}{dx^2} = - \left(\frac{2\pi}{\lambda} \right)^2 \psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} = -\left(\frac{2\pi}{\lambda}\right)^2 \psi(x)$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) \psi(x, y, z) = -\left(\frac{2\pi}{\lambda}\right)^2 \psi(x, y, z)$$

$$\nabla^2 \psi(x, y, z) = -\left(\frac{2\pi}{\lambda}\right)^2 \psi(x, y, z)$$

The combination of second partial derivatives is called Laplacian

For a particle with Energy E in a region of potential energy V

$$\nabla^2 \psi(x, y, z) = -\left(\frac{2\pi}{\lambda}\right)^2 \psi(x, y, z)$$

$$\lambda = \frac{h}{[2m(E - V)]^{1/2}}$$

$$\left(\frac{2\pi}{\lambda}\right)^2 = \frac{4\pi^2 \times 2m \times (E - V)}{h^2} = \frac{8\pi^2 m(E - V)}{h^2}$$

$$E = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V$$

$$p = [2m(E - V)]^{1/2}$$

$$\lambda = \frac{h}{p} = \frac{h}{[2m(E - V)]^{1/2}}$$

$$-\left(\frac{h^2}{8\pi^2m}\right)\nabla^2\psi(x,y,z)+V(x,y,z)\psi(x,y,z)=E\psi(x,y,z)$$

$$-\left(\frac{h^2}{8\pi^2m}\right)\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi(x,y,z)+V(x,y,z)\psi(x,y,z)=E\psi(x,y,z)$$

Schrödinger Equation

- Energies of different energy levels can be obtained by solving wave equation also known as “**Schrödinger equation**”.

$$-\left(\frac{h^2}{8\pi^2m}\right) \frac{d^2\psi(x)}{dx^2} + V \psi(x) = E \psi(x)$$

$$\hat{H} \psi(x) = E \psi(x)$$

The solution of this equation can give wavefunctions and energy associated with different levels.

Wavefunction

- Quantum mechanics acknowledges the wave-particle duality of matter by supposing that, rather than traveling along a definite path, a particle is distributed through space like a wave. The wave that in quantum mechanics replaces the classical concept of particle trajectory is called a wavefunction, ψ (“psi”).

A **wave function** in [quantum mechanics](#) describes the [quantum state](#) of an isolated system of one or more particles. There is *one* wave function containing all the information about the entire system, not a separate wave function for each particle in the system.

Postulates of Quantum Mechanics

Postulate 1:

State and wave functions. Born interpretation

Meaning of wave function:

$$P(\mathbf{r}) = |\psi|^2 = \int \psi^* \psi d\tau$$

=> the probability that the particle can be found at a particular point $\mathbf{x}$ and a particular time t . (Born's / Copenhagen interpretation)

Implications of Born's Interpretation

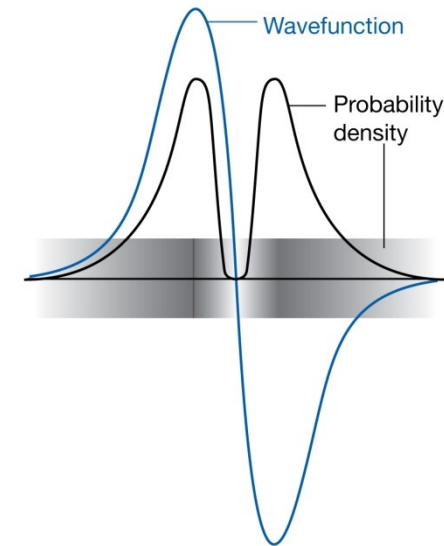
(1) Positivity:

$$P(\mathbf{r}) \geq 0$$

The sign of a wavefunction has no direct physical significance:
The positive and negative regions of this wavefunction both correspond to the same probability distribution.

(2) Normalization:

$$\int_{all_space} \Psi^* \Psi d\tau = 1$$



i.e. the probability of finding the particle in the universe is 1.

Physically acceptable wave function

- The wave function and its first derivative must be:
 - 1) Finite. The wave function must be single valued. This means that for any given values of x and t , $\Psi(x,t)$ must have a unique value. This is a way of guaranteeing that there is only a single value for the probability of the system being in a given state.

2. Square-integrable

The wave function must be square-integratable. In other words, the integral of $|\Psi|^2$ over all space must be finite. This is another way of saying that it must be possible to use $|\Psi|^2$ as a probability density, since any probability density must integrate over all space to give a value of 1, which is clearly not possible if the integral of $|\Psi|^2$ is infinite. One consequence of this proposal is that Ψ must tend to 0 for infinite distances.

$$\int_{-\infty}^{+\infty} \psi^* \psi d\tau = 1$$

Normalization

$$\int \psi^* \psi d\tau = A$$

$$\int_{-\infty}^{+\infty} \frac{\psi^*}{\sqrt{A}} \frac{\psi}{\sqrt{A}} d\tau = 1$$

$$\int_{-\infty}^{+\infty} \psi^* \psi d\tau = \infty$$

$$\int_{-\infty}^{+\infty} \frac{\psi^*}{\sqrt{\infty}} \frac{\psi}{\sqrt{\infty}} d\tau = A < \infty$$

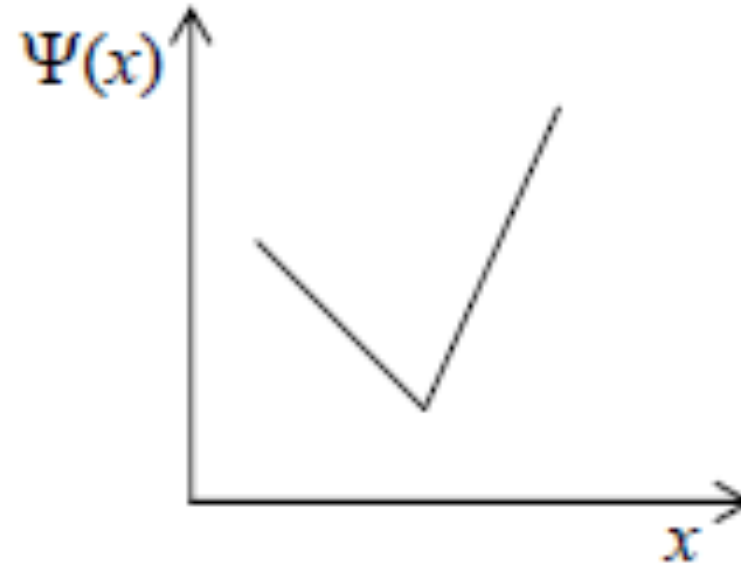
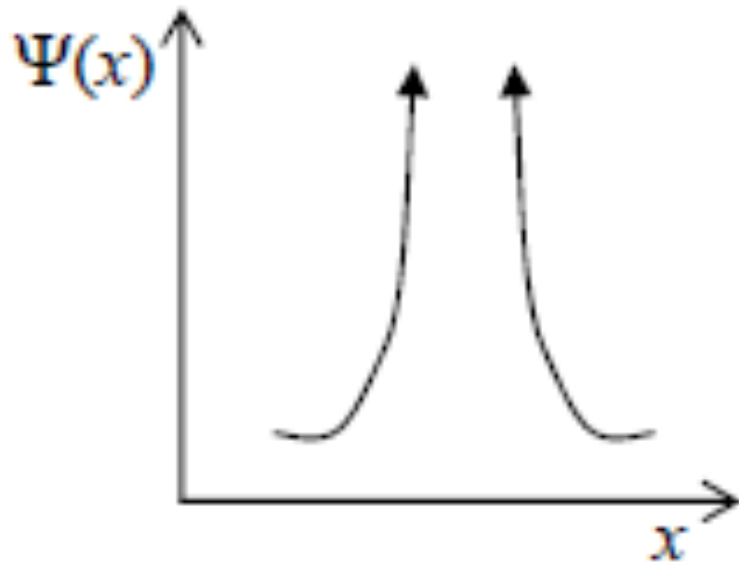
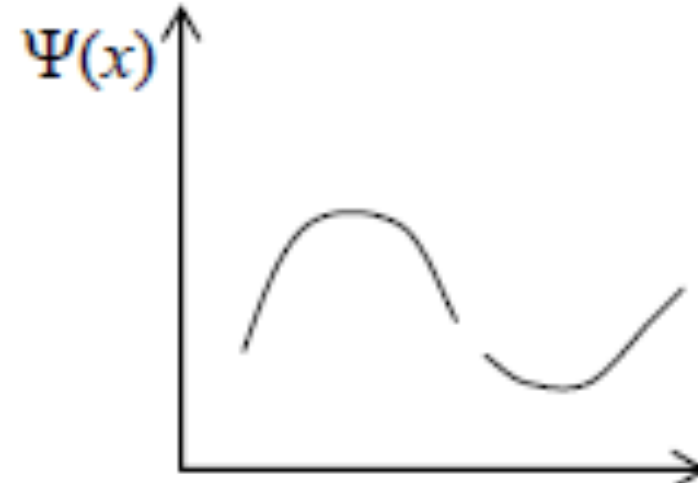
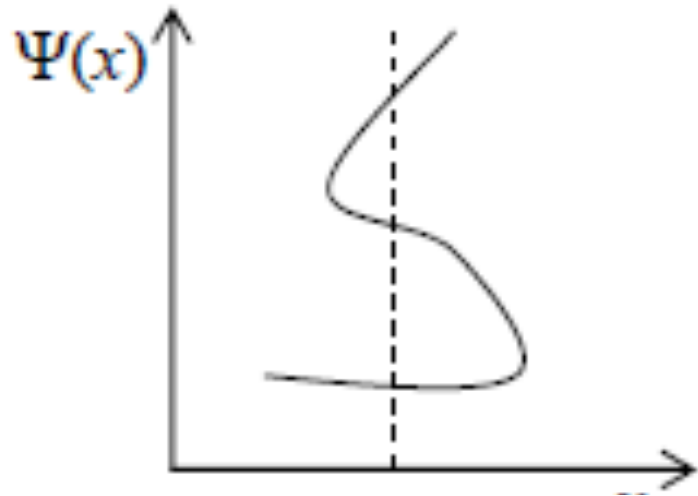
Continuous wavefunction

- A rapid change would mean that the derivative of the function was very large (either a very large positive or negative number). In the limit of a step function, this would imply an infinite derivative. Since the momentum of the system is found using the [momentum operator](#), which is a first order derivative, this would imply an infinite momentum, which is not possible in a physically realistic system.

Continuous First derivative

1. All first-order derivatives of the wave function must be continuous. Following the same reasoning as in condition 3, a discontinuous first derivative would imply an infinite second derivative, and since the [energy](#) of the system is found using the second derivative, a discontinuous first derivative would imply an infinite energy, which again is not physically realistic.

Acceptable or Not ??



Acceptable or not acceptable ??

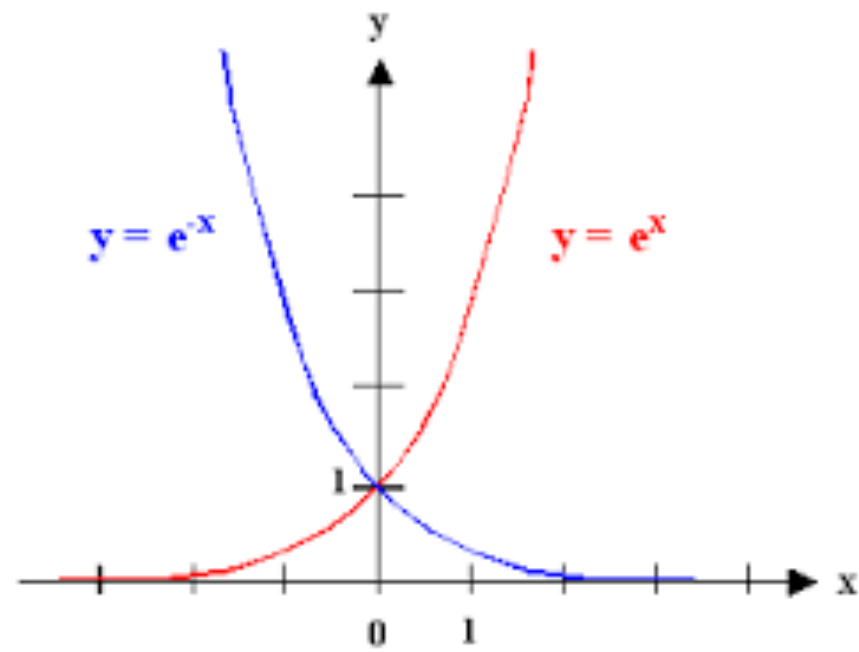
$$(i) e^{-x} (0, \infty)$$

$$(ii) e^{-x} (-\infty, \infty)$$

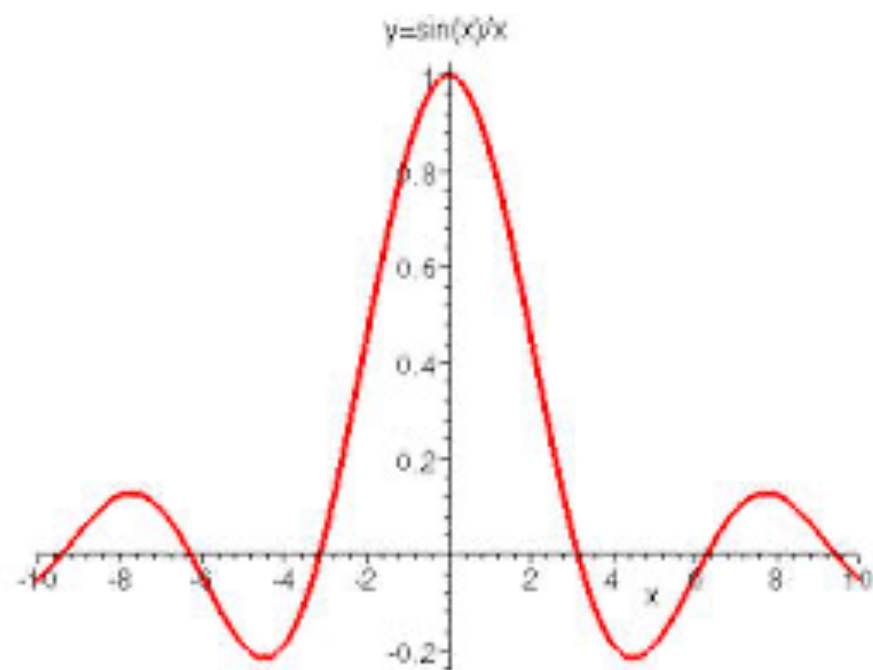
$$(iii) \frac{\sin x}{x}$$

$$(iv) \sin^{-1} x$$

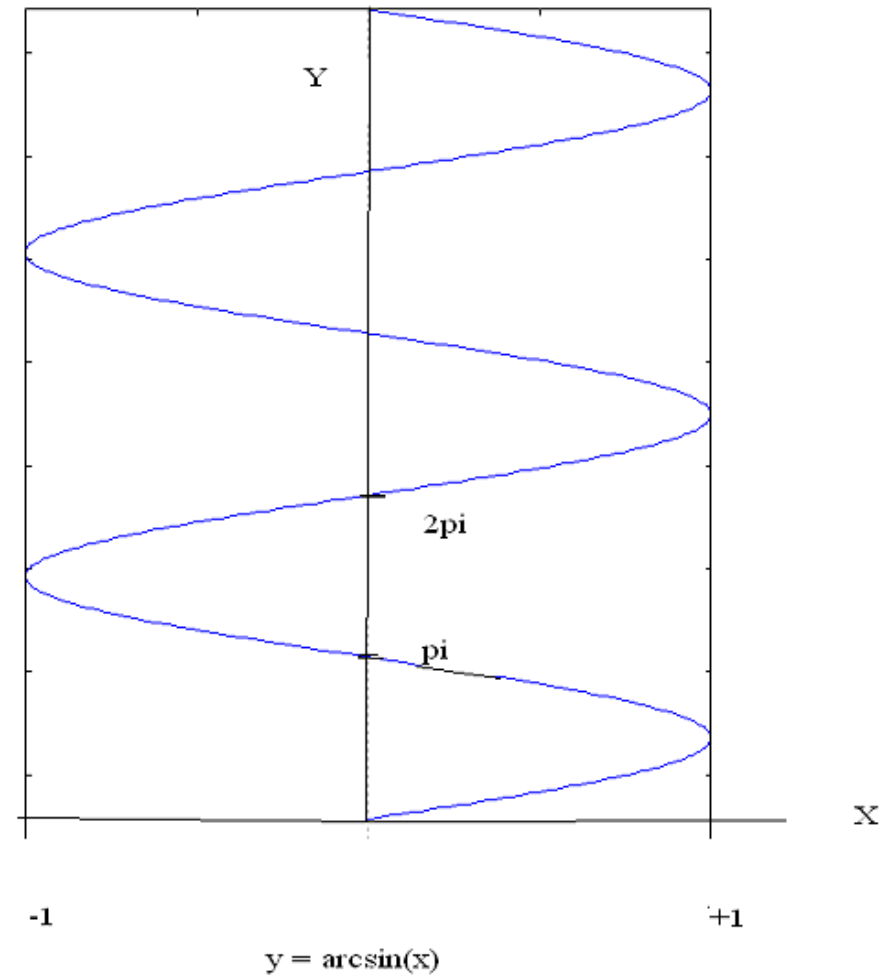
Exp(x)



$$\text{Sinx}/x$$



$$\text{Sin}^{-1}x$$



Not acceptable function